



FACULTY OF INFORMATION SCIENCE

UNIVERSITI TEKNOLOGI MARA

BACHELOR OF INFORMATION SCIENCE (HONS.)

INFORMATION SYSTEM MANAGEMENT (SI262)

ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT (IMS566)

GROUP ASSIGNMENT:

BLOOD DONATION APPOINTMENT AND DONOR MANAGEMENT

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17 JULY 2026

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ACKNOWLEDGEMENT

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

We would like to begin by expressing our sincere appreciation to God for His blessings and guidance that made this project possible. Our deepest gratitude goes to Dr. Muhammad Asyraf Binti Wahi Anuar, our lecturer for subject Advanced Web Design and Content Management (IMS566), whose continuous assistance, insightful feedback and valuable knowledge greatly contributed to the successful completion of this project.

We would also like to extend our thanks to our friends and peers for their camaraderie by offering constructive comments and providing encouragement along the way. Their support created positive environment that strengthened our degree's experience.

Lastly, we are truly grateful to our family for their constant prayers, motivation and unwavering moral support. Their presence and encouragement have always been a source of strength and for that we are deeply thankful for everything they have done.

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1.0 Introduction

The Blood Donation Appointment and Donor Management System is a web-based application developed to improve the efficiency of managing blood donation activities. The system allows donors to register, schedule blood donation appointments, and view their donation history through a centralized platform. It also enables staff to manage donor information, verify appointments, maintain donation records, and monitor blood donation activities in an organized and systematic manner.

This project was developed using CakePHP, an open-source PHP web application framework designed to simplify and accelerate web application development. CakePHP provides various built-in features such as database integration, authentication, form validation, routing, and CRUD (Create, Read, Update, and Delete) operations, allowing developers to build secure, reliable, and maintainable applications with less coding effort. Its structured development approach also improves code organization and makes future system enhancements easier to implement.

The system uses MySQL as its relational database management system to store and manage data related to donors, blood types, appointments, staff, and donation histories. The database is designed based on an Entity-Relationship Diagram (ERD) to establish relationships between entities and ensure data consistency, integrity, and efficient data retrieval.

Overall, the Blood Donation Appointment and Donor Management System provides a digital solution that replaces manual record management with a more efficient and reliable system. By integrating donor registration, appointment scheduling, staff management, blood type records, and donation history into a single platform, the system enhances operational efficiency, improves data accuracy, and supports better management of blood donation services.

2.0 GitHub Repository Link

The complete source code for the Blood Donation Appointment and Donor Management System is maintained using GitHub as the version control platform. GitHub allows team members to collaborate efficiently by tracking code changes, managing different project versions, and maintaining a centralized repository throughout the development process.

Repository Link:

<https://github.com/mahyyrah/redconnect-2.git>

3.0 Entity-Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the logical structure of the Blood Donation Appointment and Donor Management System database. The database consists of five entities that are connected through primary and foreign key relationships to maintain data integrity and reduce redundancy.

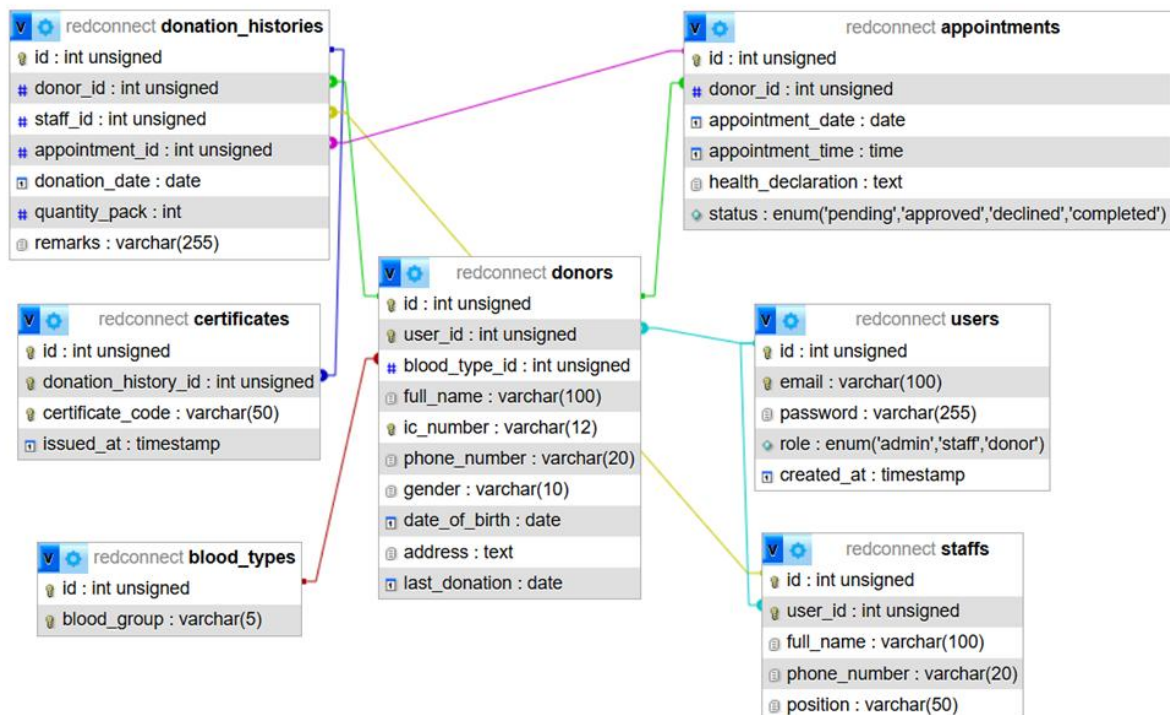


Figure 1: RedConnect ERD

4.0 System Requirements

4.1 Software

The following software is required to run the RedConnect system:

CATEGORY	SOFTWARE	DESCRIPTION
Operating sytem	Windows 11	The setting for development and deployment
Web server stack	Laragon	Local development environment containing Apache, PHP and MySQL
Database server	MySQL/MariaDB	To store and manage system data
PHP Runtime	PHP 8.1/8.2/8.3	To execute the CakePHP application source data
Package manager	Composer	PHP dependency manager to install CakePHP and its libraries
Web browser	Google Chrome	To test, view and debug the user interface
Text editor	Visual Studio Code	For development purposes



Figure 2: Laragon running with Apache and MySQL started

4.2 Version compatibility

The Red Connect system has been tested and verified to work with the following software versions:

COMPONENT	MINIMUM VERSION	RECOMMENDED VERSION
PHP	7.4	8.1 and higher
CakePHP Framework	4.5.0	4.5.0
MySQL	5.7	8.0
MariaDB	10.3	10.6
Composer	2.0	2.6 and higher
Apache	2.4	2.4

The Red Connect system uses PHP 7.4 as the minimum requirement due to CakePHP 4.x dependencies. However, PHP 8.x is fully supported and recommended for better performance and security.

4.3 Dependencies

The Red Connect system relies on the following third-party libraries and packages:

4.3.1 Core Dependencies

PACKAGE	VERSION	PURPOSE
Cakephp/cakephp	4.5.0	Core CakePHP framework for application development using the MVC architecture.
Cakephp/authentication	3.11	Provides user authentication features such as login, logout, and identity management.
Cakephp/migrations	3.7	Manages database schema changes and version control through migrations.
Cakephp/plugin-installer	1.3	Automatically installs and configures CakePHP plugins by composer
Mobiledetect/mobiledetectlib	3.74	Detects mobile, tablet, and desktop devices to support responsive application behaviour.

4.3.2 Development Dependencies

PACKAGE	VERSION	PURPOSE
Cakephp/bake	2.8	Generates CakePHP application code such as controllers, models, templates and CRUD scaffolding to speed up development.
Cakephp/debug_kit	4.9	Provides debugging and profiling tools, including request information, SQL queries, logs and performance analysis during development.
Phpunit/phpunit	9.6	Executes automated unit and integration tests to verify application functionality and ensure code quality.
Cakephp/cakephp-codesniffer	4.5	Checks source code against CakePHP coding standards to maintain consistent code style and quality.

5.0 User Interface Overview

RedConnect's website features a sleek and modern design. The interface is designed to be user friendly for everyone who wants to donate blood for a good cause. The interface can be used on the desktop.

5.1 Overall Layout Structure

When logging into the website, the system has a consistent layout throughout most of the pages with the following components:

1. **Logo:** The Logo of the website permanently sits at the upper left corner of the screen to remind users what website they are on.
2. **Header:** The Header sits at the top of the screen. It displays what page the user is on, a description of said page, and customizable features for the user's convenience
3. **Sidebar Navigation:** At the left side of the screen are the various buttons to help users navigate throughout the website.
4. **Main Content Area:** In the middle of the screen is the content of the website. The content will change depending on the pages users choose to be in.

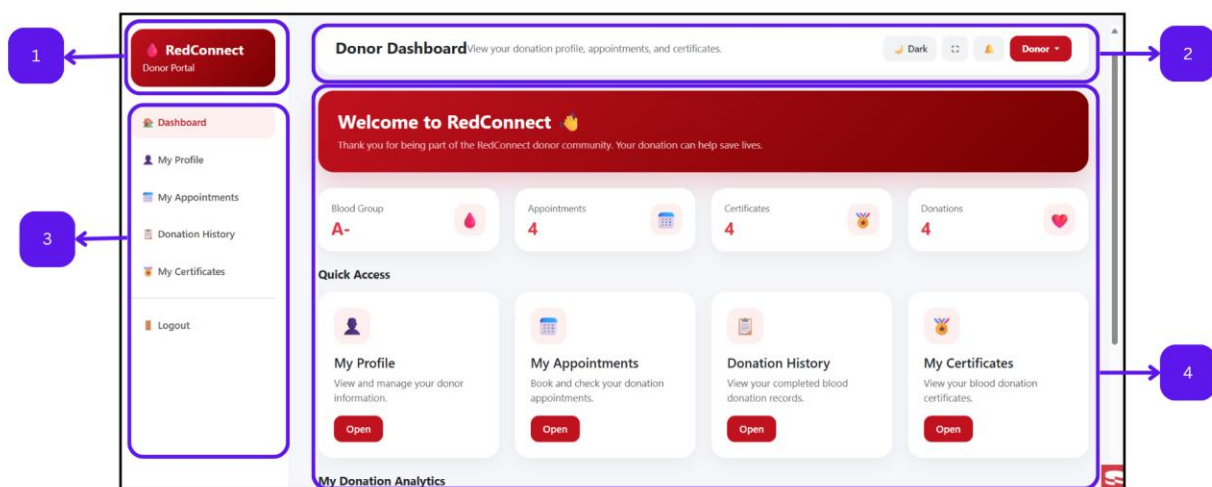


Figure: 3 Overall Layout Structure

5.2 Colour Scheme & Branding

RedConnect has a signature branding that the website follows:

Element	Colour	Hex Code
Primary Colour	Red	#C1121F
Gradient	Red to Dark Red	#BF111E - #7E0102
Accent Colour	Light Peach	#FFF0F0
Background Page	Light Grey	#F6F7FB
Textbox	White	#FFFFFF
Text	Dark Grey	#343339

5.3 Navigation System

The navigation system changes to adapt to the two different user's role, donor and staff.

Donor's Navigation Menu	Staff's Navigation Menu
• Dashboard • My Profile • My Appointments • Donation History • My Certificates • Logout	• Dashboard • Users • Donors • Staffs • Blood Types • Appointments • Donation History • Certificates • Logout

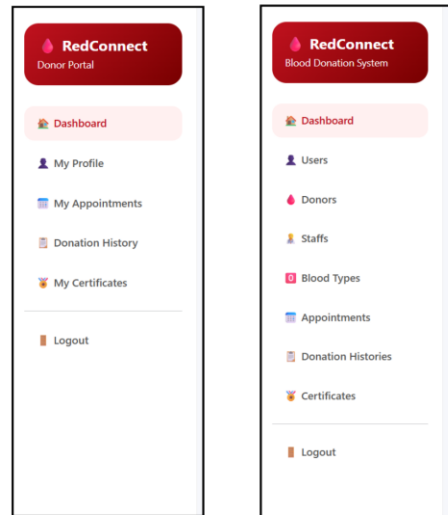


Figure 4: Navigation Menu for Donors (Left) and Navigation Menu for Staff (Right)

6.0 Features & Functionalities

6.1 Logging In

6.1.1 Login Page

Before accessing the site, users will be met with a Login page where they must input their email and password. Logging into the website is different for regular users (donors) and staff members.

Donors can access the site by inputting their personal email and password. For staff members, they are required to input their RedConnect email instead to access the staff site. The RedConnect email is different to a regular email, the address being *[staff name]@redconnect.mail.com*.

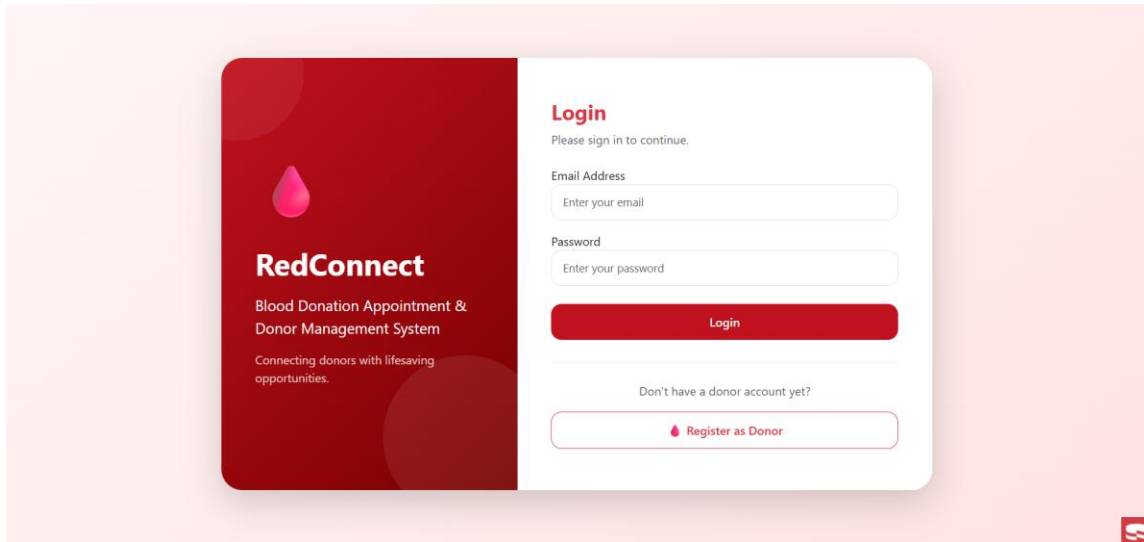


Figure 5: Login Page

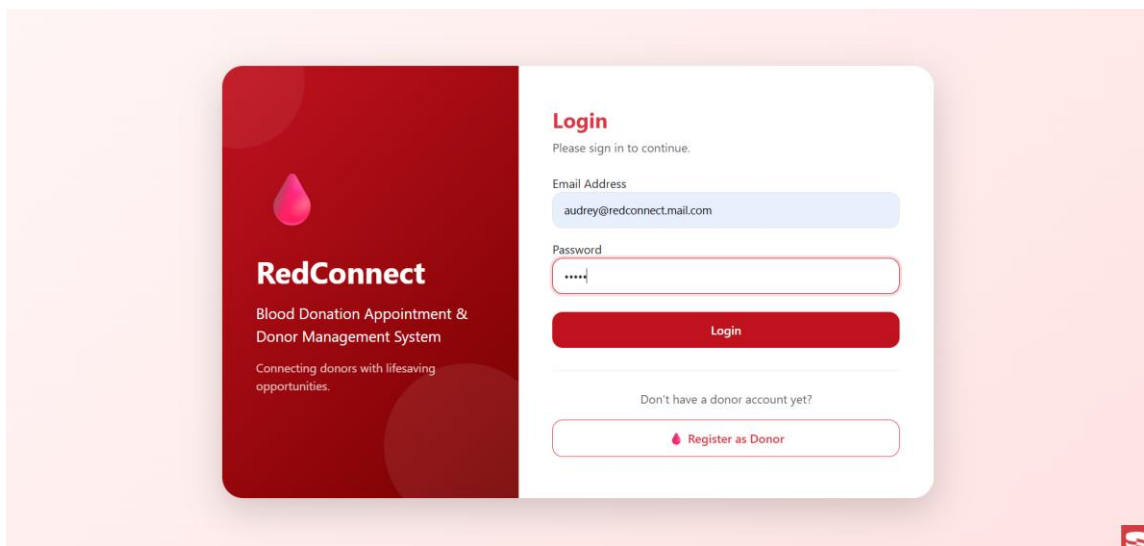


Figure 6: Login Page for Staff

6.1.2 Registration Page

Donors must register on the website first. Donors can do so by clicking on the 'Register as Donor' button at the bottom. The registration page requires personal information from the donor which are:

Login Information:

- Email Address
- Password

Donor Profile:

- Full Name
- IC Number
- Phone Number
- Blood Type
- Gender
- Date Of Birth
- Last Donation Date
- Address

Create Donor Account
Register as a blood donor in RedConnect.

Login Information

Email Address

Password

Donor Profile

Full Name

IC Number

Phone Number

Blood Type

Gender

Date of Birth

Last Donation Date

Address

Figure 7: Registration Page

After registration, donors can access the site using the same login information they use when registering.

6.2 Web Content for Donors

6.2.1 Dashboard

Dashboard is the homepage of RedConnect. Users will be met with this page when logging in. The dashboard displays:

- A Welcome banner
- User information including blood group, Appointments, Certificates, and Donations
- Quick Access to My Profile, My Appointments, Donation History, and My Certificates
- User's Donation Analytics

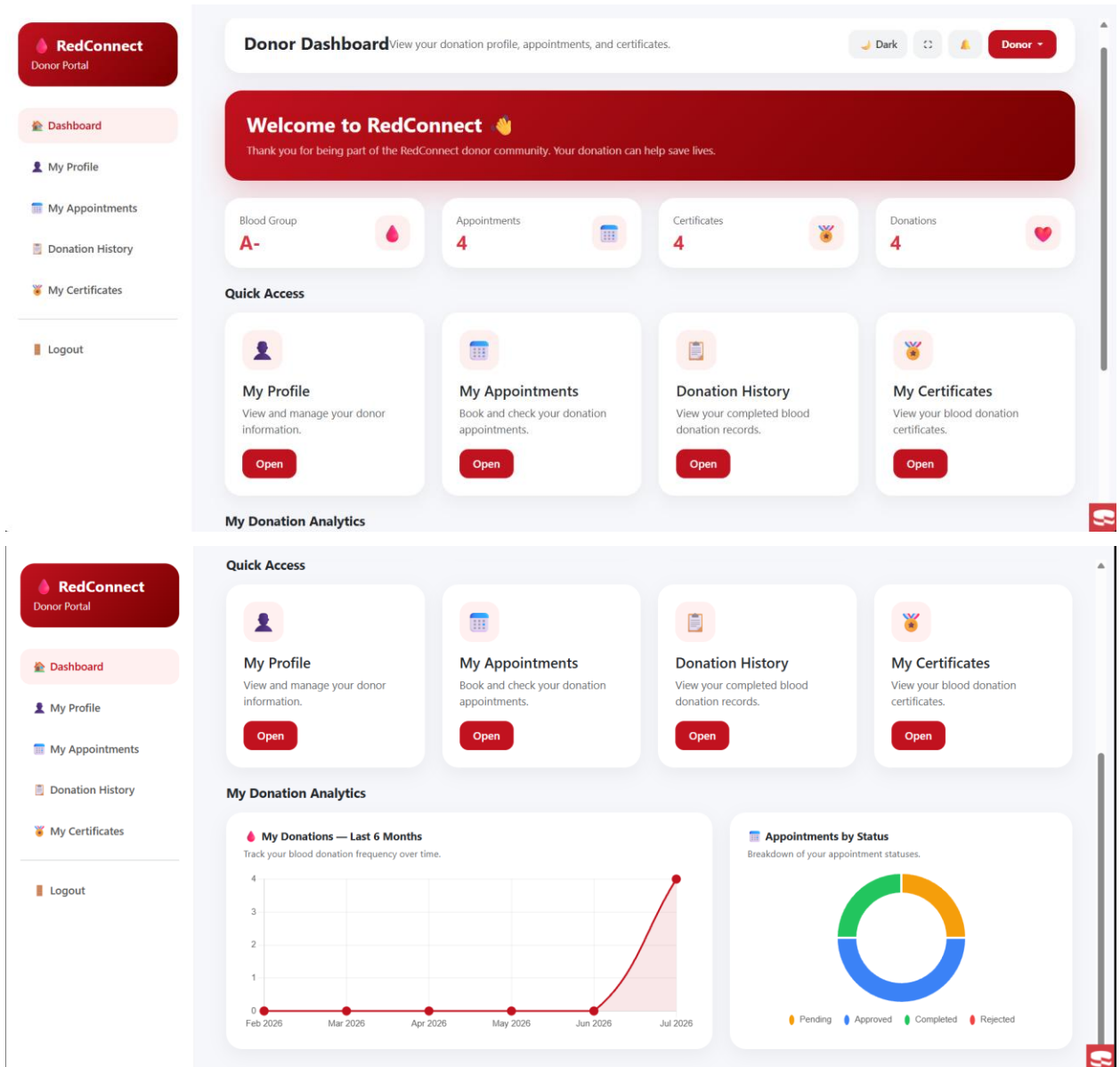


Figure 8: Dashboard for Donors

6.2.2 My Profile

Users can view their profile page by using the quick access button or the sidebar navigation. The page will display the information users has input during registration which are:

- Full name
- Email Address
- Blood Group
- IC Number
- Phone Number
- Gender
- Date of Birth
- Last Donation
- Address

RedConnect
Donor Portal

My Profile View your donor information and personal details

Dark [icon] [icon] [icon] Donor

Gloria Audrey
RedConnect Blood Donor

Email Address
audreygloria26@gmail.com

Blood Group
A-

IC Number
040909131122

Phone Number
0178038462

Gender
Female

Date of Birth
9/9/04

Last Donation
7/23/26

Address
Pangsapuri Resak

Update Profile

Figure 9: My Profile

If there is information donors would like to change or update, they can click on the 'Update Profile' button at the bottom. Doing so will let donors change certain information such as phone number or address. Donors can save changes or cancel editing.

The screenshot shows the 'Update Profile' interface in the RedConnect Donor Portal. On the left is a sidebar with navigation links: Dashboard, My Profile (highlighted), My Appointments, My Certificates, and Logout. The main content area is titled 'Update Profile' with a subtitle 'Update your phone number and address.' and a close button. A red header bar displays the user's name 'Gloria Audrey' and a note 'Edit allowed details only'. Below this, various fields are shown: Email Address (audreygloria26@gmail.com), Blood Group (A-), IC Number (040909131122), Gender (Female), Date of Birth (9/9/04), Phone Number (0178038462), Last Donation (7/23/26), and Address (Pangsapuri Resak). At the bottom right are 'Cancel' and 'Save Changes' buttons.

Field	Value
Email Address	audreygloria26@gmail.com
Blood Group	A-
IC Number	040909131122
Gender	Female
Date of Birth	9/9/04
Phone Number	0178038462
Last Donation	7/23/26
Address	Pangsapuri Resak

Figure 10: Editing My Profile

6.2.3 My Appointments

This page allows donors to book an appointment and view their appointment status. Since RedConnect's goal is to receive blood donations, this page is essential. Donors can book an appointment by choosing a date, picking a time slot, and providing their health declaration in the respective boxes. Click on the 'Submit Application' to officially book an appointment.

At the bottom of the booking menu, it allows donors to see their booking history. Donors must keep in watch of the status indicator to see if their appointment is approved by staff. Before approval, it will be displayed as 'Pending'.

No.	Date	Time	Health Declaration	Status
1	7/24/26	12:39 AM	I am healthy	Pending
2	7/23/26	12:47 AM	I am healthy	Approved

Figure 11: My Appointments

6.2.4 Donation History

This page allows users to view their confirmed blood donations. The page displays:

- number of donations
- donation dates
- quantity pack
- staff member who helped with the donation
- the time the donation takes place
- remarks regarding the donation
- view of certificates

RedConnect
Donor Portal

Dashboard

My Profile

My Appointments

Donation History

My Certificates

Logout

Donation History View your completed blood donations. X

Dark

Donor

Completed Donations
These records are created after staff confirm your blood donation.

No.	Donation Date	Quantity Pack	Staff	Appointment	Remarks	Certificate
1	7/23/26	1 Pack	Zulaikha Binti Mohd Khairul Sabastian	7/11/26 10:30 AM	-	View
2	7/7/26	1 Pack	Zulaikha Binti Mohd Khairul Sabastian	7/11/26 10:30 AM	-	View
3	7/7/26	1 Pack	Zulaikha Binti Mohd Khairul Sabastian	7/11/26 10:30 AM	-	View
4	7/7/26	2 Pack	Zulaikha Binti Mohd Khairul Sabastian	7/11/26 10:30 AM	-	View

Figure 12: Donation History

6.2.5 My Certificates

This page allows users to view and print their generated certificates after they successfully complete their blood donations. It will display:

- The number of Donations
- Certificate Code
- Donor Name
- Blood Group
- Donation Date
- Time of Issue
- Button to View Certificate

Clicking on the 'View' button will lead donors to a page for them to fully view the certificate. Donors have the option to print the certificate or save it as a PDF file.

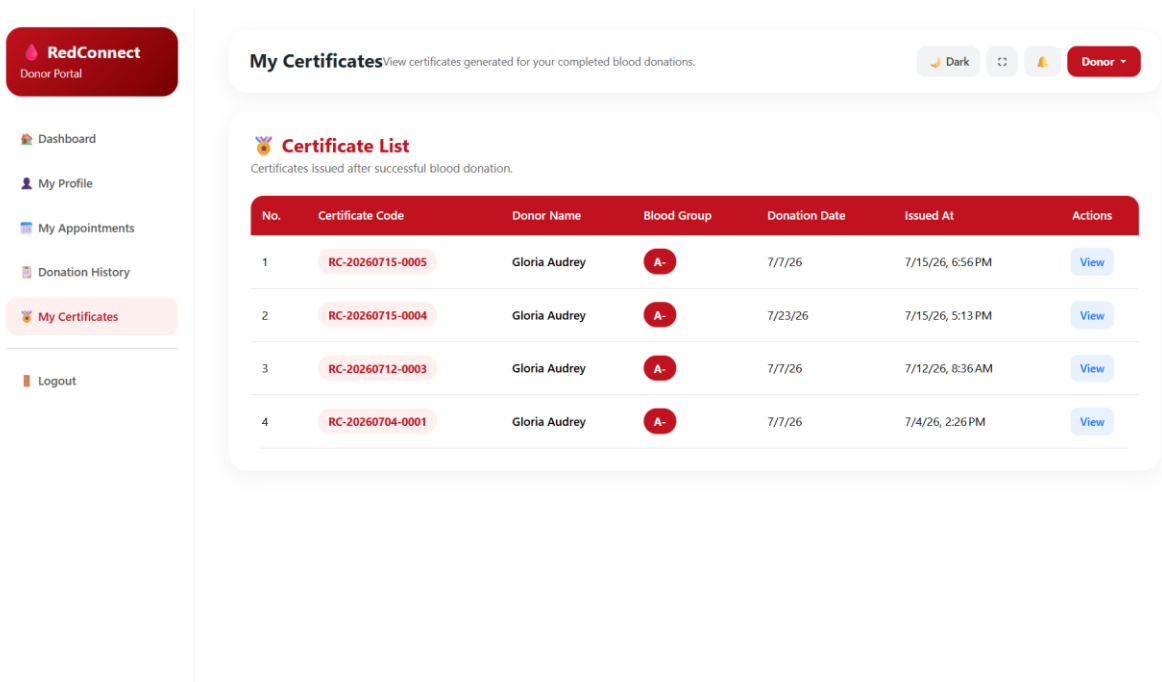


Figure 13: My Certificate

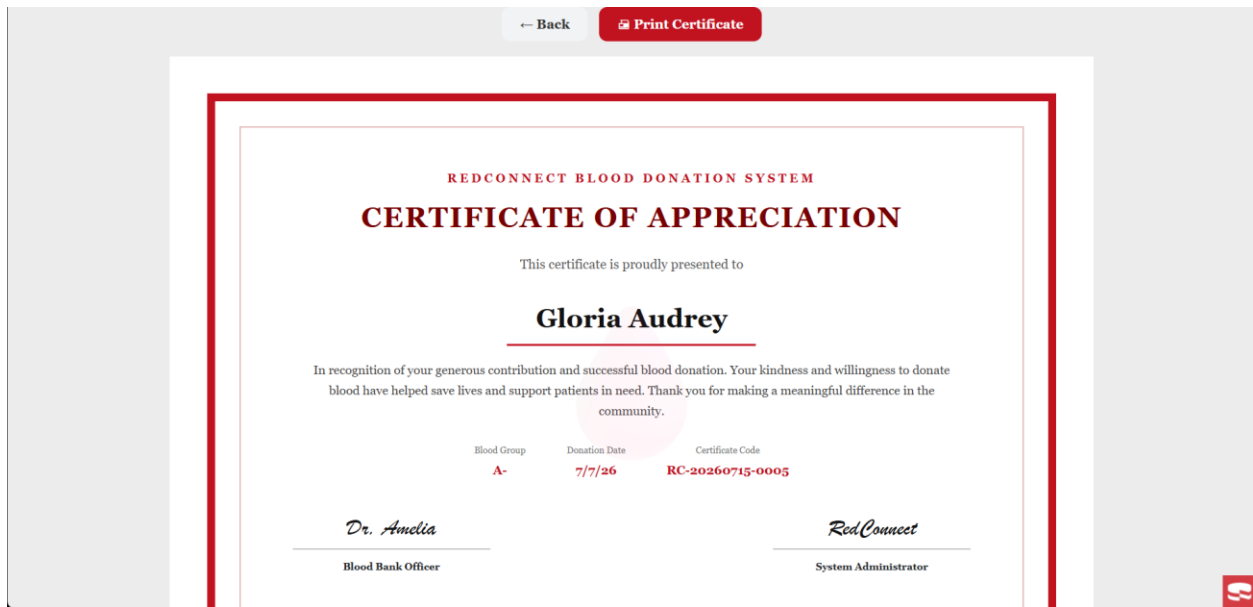


Figure 14: Certificate

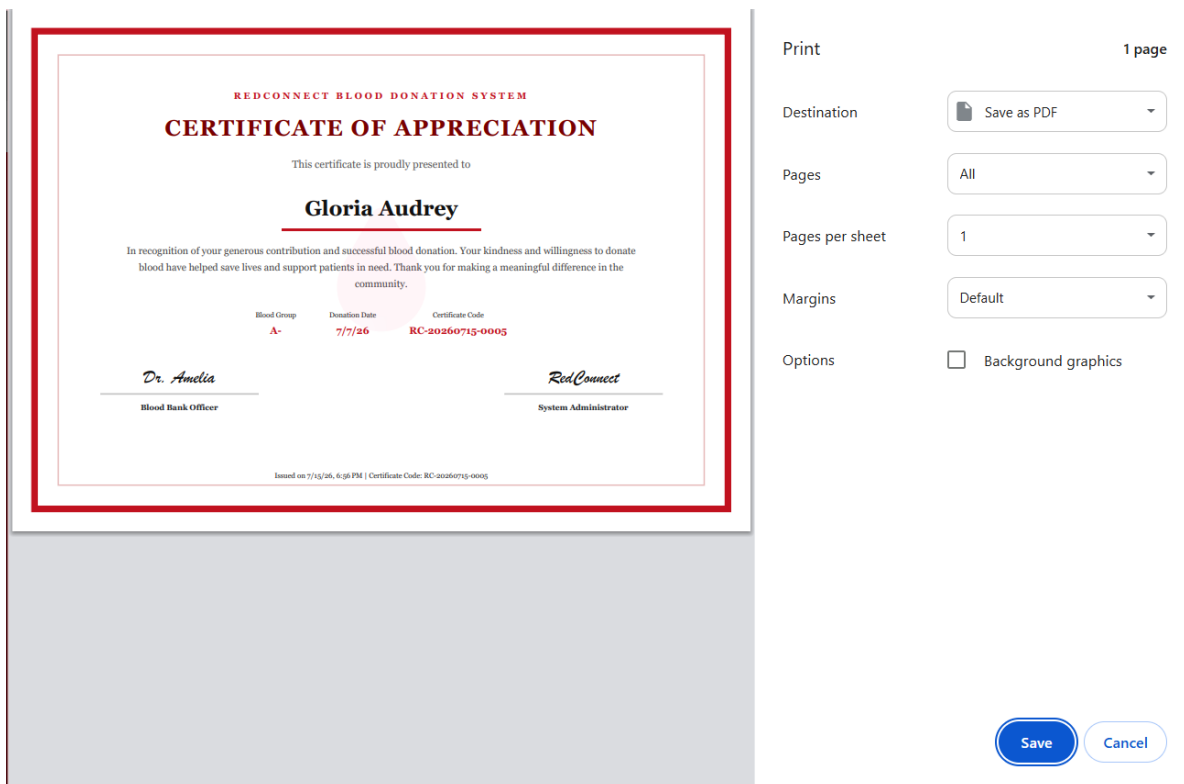


Figure 15: Printing The Certificate

6.3 Web Content for Staff

6.3.1 Dashboard

Similar to the donors' experience, logging into the site as a staff member will bring staff to their own dashboard page. From here, staff can manage the system easily. The page displays:

Database Information:

- Amount of Blood Groups
- Amount of Staffs
- Amount of Donors
- Amount of Donation Records

Quick Access to System Modules:

- Users
- Donors
- Staffs
- Blood Types
- Appointments
- Donation History
- Certificates

System Analytics:

- Donations by Blood Group
- Monthly Donation Trend

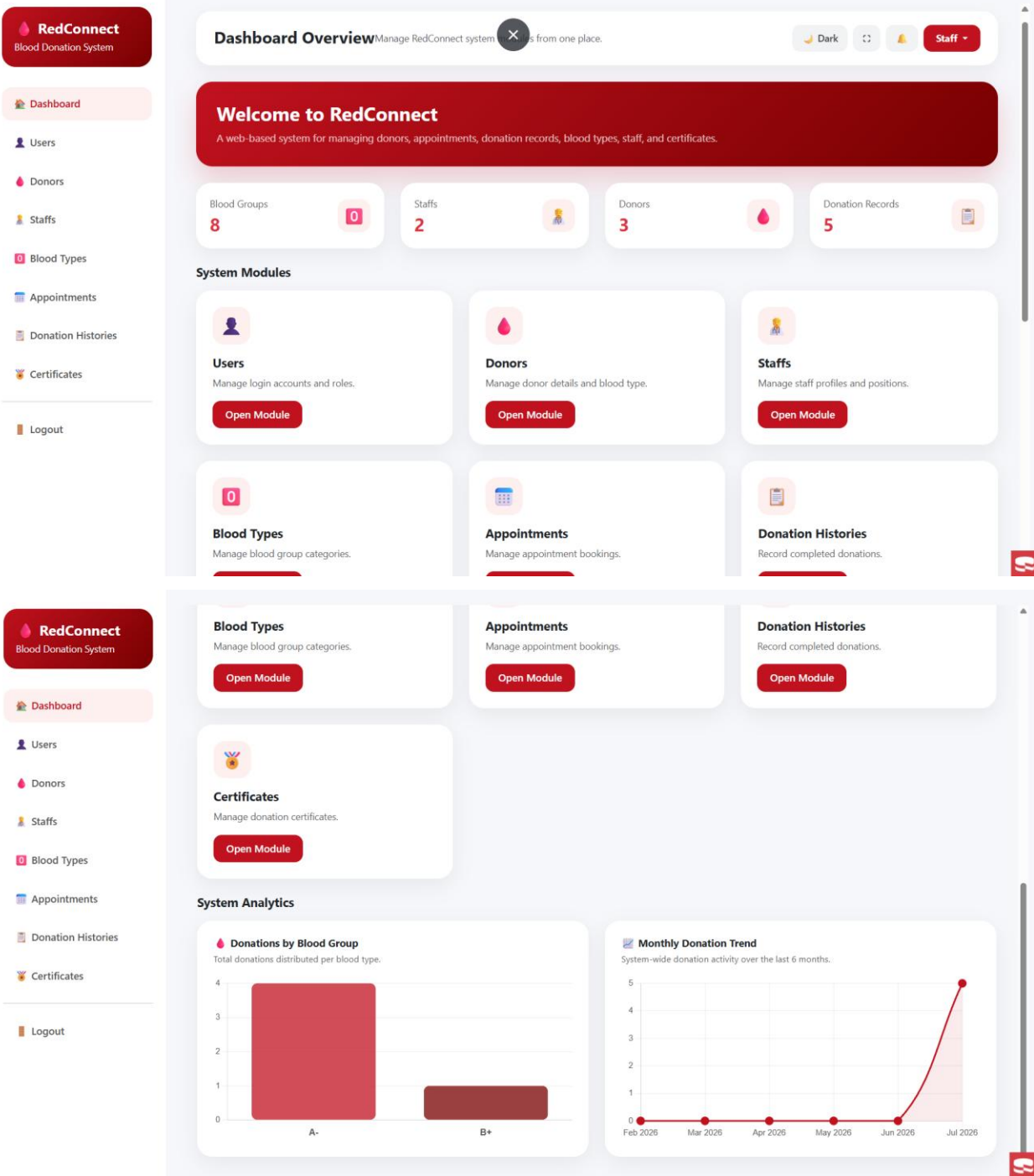


Figure 16: Dashboard for Staff

6.3.2 Users

The User's page is for staff members to manage login accounts and system roles of the system. The page will display a list of all the users that are registered into the system with the email of the users, role in the system, date of creation, and actions the staff member can take on the users. Staff can view, edit, or delete existing registered users. Staff can also add a new user as well.

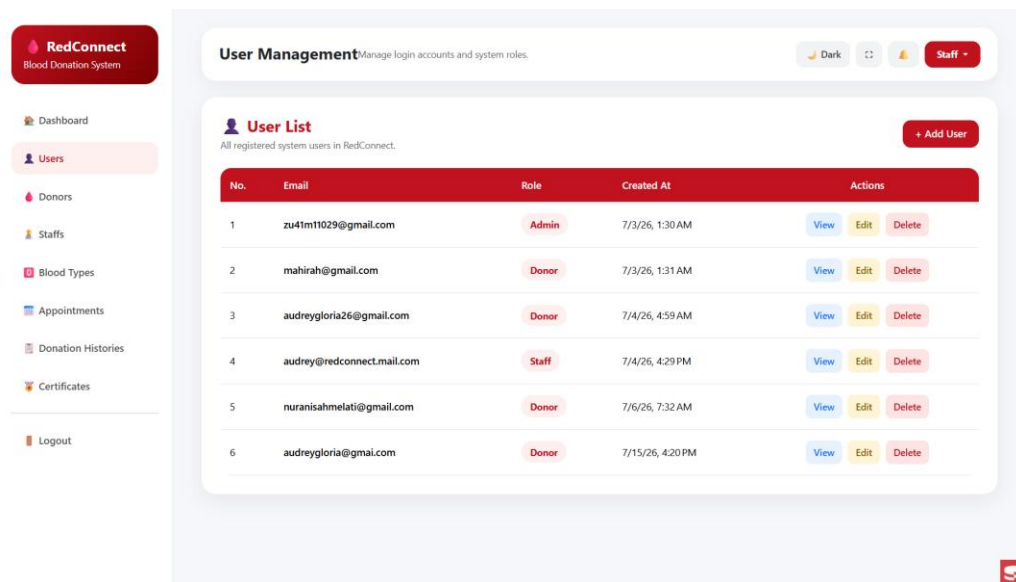


Figure 17: User List

Clicking on 'View' user will lead staff to a User Details page where it displays the User ID, Email Address, User Role, and Date of Creation. From the page, staff have the option to go back to previous page, edit the User Details or Delete the user.

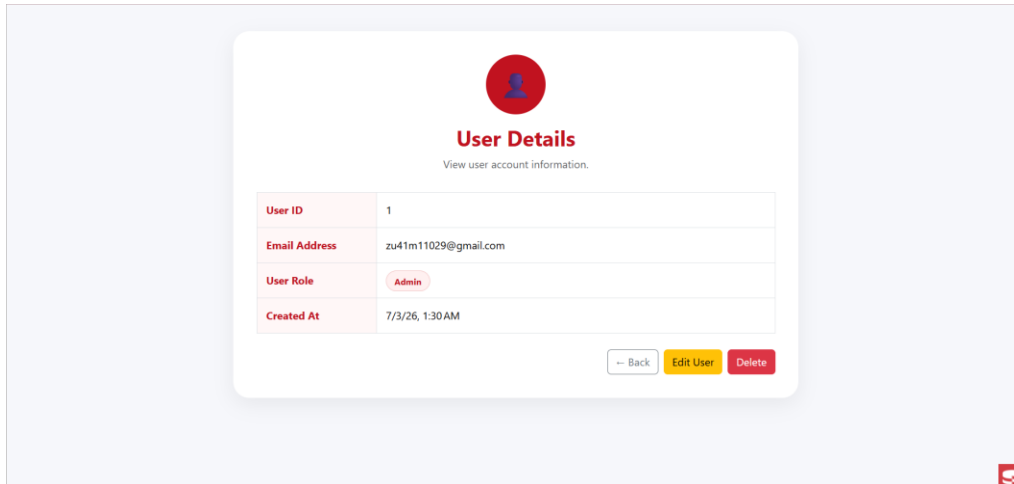


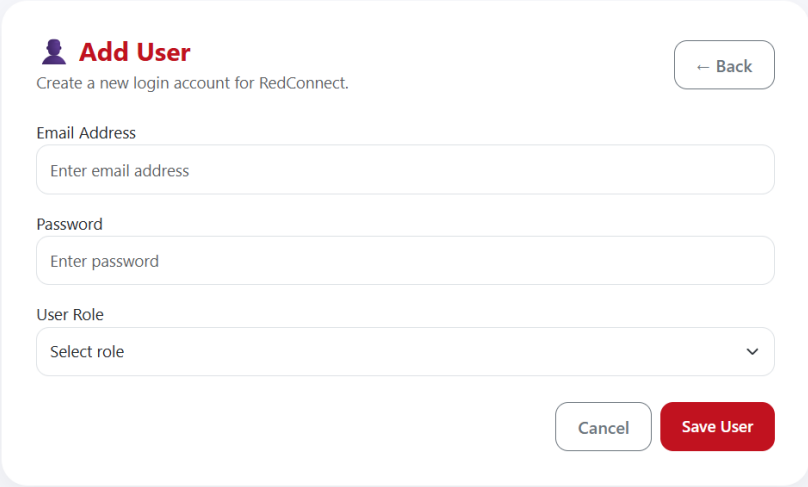
Figure 18: Viewing User

When editing a user's details, staff can edit the email address, password, and user role. To confirm the edit, staff must click on the 'Update User' button. If not, staff can also cancel the edit. Staff also have the option to delete user on the page.

A screenshot of a web application interface titled "Edit User" with a subtitle "Update user account details." The page features a red pencil icon next to the title. At the top right is a "Back" button. The form contains three input fields: "Email Address" with the value "zu41m11029@gmail.com", "Password" with masked characters "*****", and "User Role" with a dropdown menu showing "Admin". At the bottom are three buttons: "Delete User" (red outline), "Cancel" (grey), and "Update User" (red).

Figure 19: Editing User

When adding a user, staff can add the email address, password, and user role. To officially add the user, staff must click on the 'Save User' button. Clicking on 'Cancel' will cancel the action.



The screenshot shows a web form titled "Add User" with a user icon. Below the title is the instruction "Create a new login account for RedConnect." and a "← Back" button. The form contains three input fields: "Email Address" with placeholder text "Enter email address", "Password" with placeholder text "Enter password", and "User Role" with a dropdown menu showing "Select role". At the bottom right are two buttons: "Cancel" and "Save User".

Figure 20: Adding User

6.3.3 Donors

This page is where staff manages donors' personal details and blood type information. The page will display a list of registered donors to RedConnect. A search engine is provided so staff can search for something more specific. The list of donors consists of the number, donor name, email, blood group, IC number, Phone Number, Gender, Date of Last Donation, and actions that the staff members could do on the donors' profile such as view, edit, and delete. Staff can also add new donors into the system.

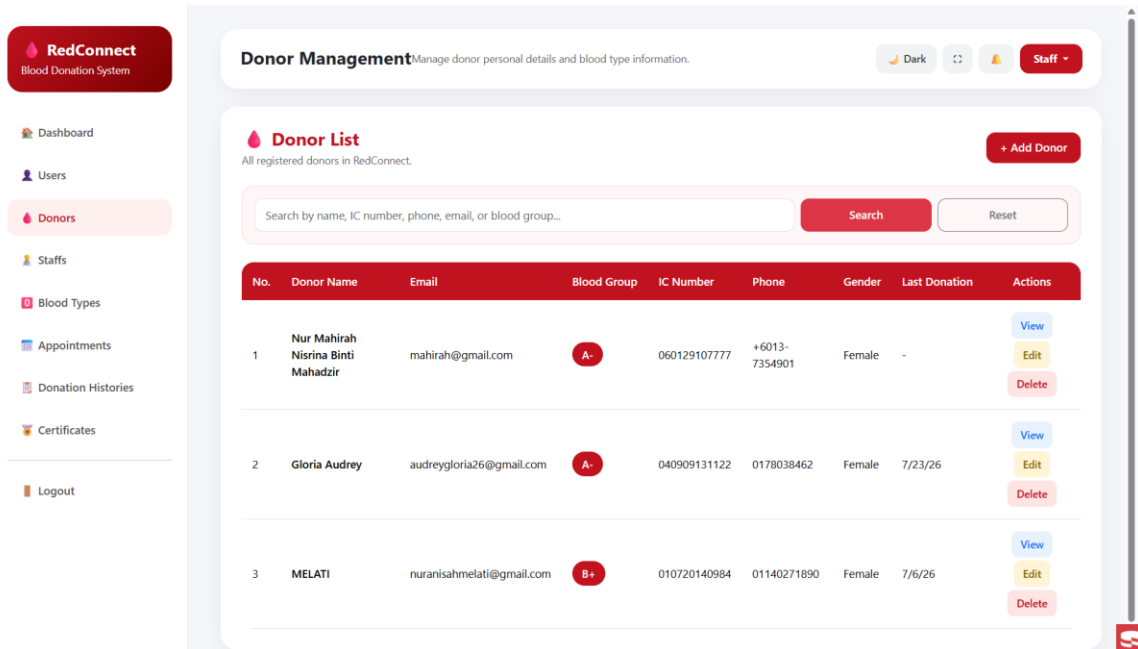


Figure 21: Donor List

When viewing a donor's profile, staff will be taken to the Donor's Details page. The page will display the donor's full name, email account, blood type, IC number, Phone Number, Gender, Date of Birth, Date of Last Donation, Address, and Appointment Records.

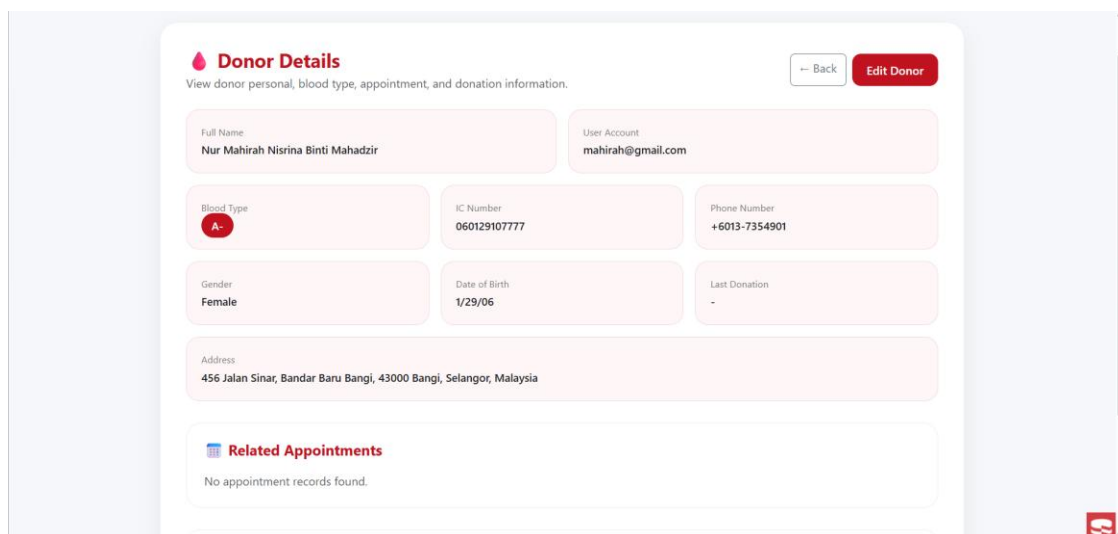
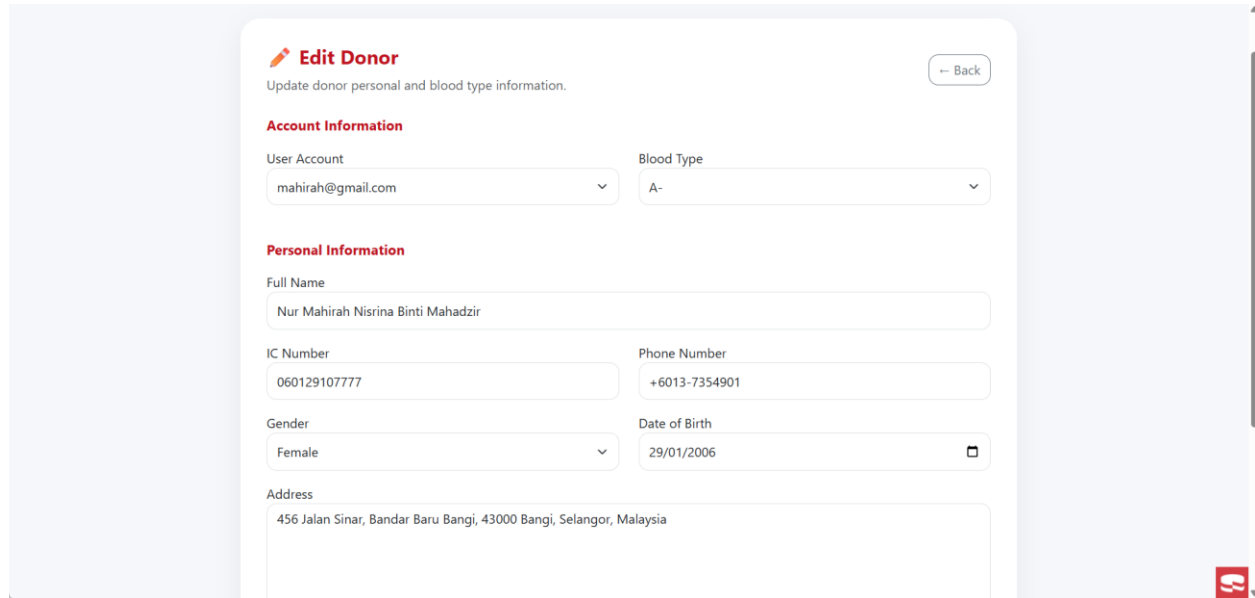


Figure 22: Viewing Donor

When editing a donor, staff can edit the account information as well as the personal information.



The screenshot displays a web interface for editing a donor's profile. At the top, a red pencil icon is followed by the title "Edit Donor". Below this, a subtitle reads "Update donor personal and blood type information." and a "← Back" button is located in the top right corner. The form is organized into two main sections: "Account Information" and "Personal Information".

Account Information

- User Account:** A dropdown menu showing "mahirah@gmail.com".
- Blood Type:** A dropdown menu showing "A-".

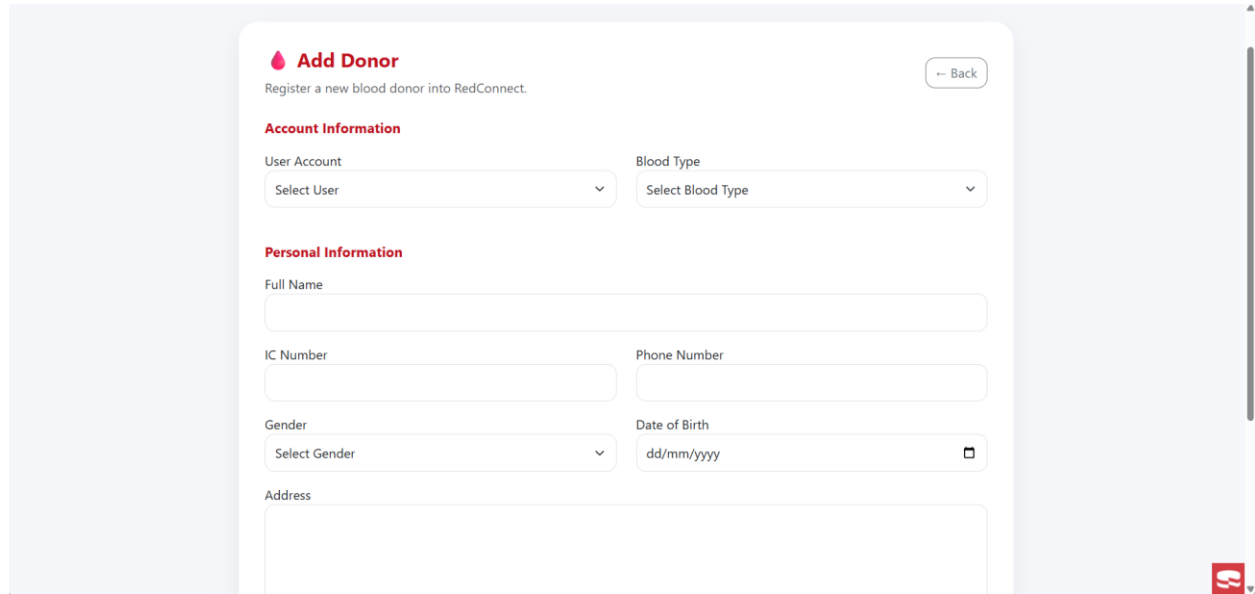
Personal Information

- Full Name:** A text field containing "Nur Mahirah Nisrina Binti Mahadzir".
- IC Number:** A text field containing "060129107777".
- Phone Number:** A text field containing "+6013-7354901".
- Gender:** A dropdown menu showing "Female".
- Date of Birth:** A date picker showing "29/01/2006".
- Address:** A text field containing "456 Jalan Sinar, Bandar Baru Bangi, 43000 Bangi, Selangor, Malaysia".

A small red logo is visible in the bottom right corner of the form area.

Figure 23: Editing Donor

When adding a new donor into the system, staff must input the account information as well as the personal information of the user.



The screenshot displays a web form titled "Add Donor" with a red blood drop icon. Below the title is the instruction "Register a new blood donor into RedConnect." and a "← Back" button. The form is organized into two main sections: "Account Information" and "Personal Information".

Account Information

- User Account:** A dropdown menu with the placeholder text "Select User".
- Blood Type:** A dropdown menu with the placeholder text "Select Blood Type".

Personal Information

- Full Name:** A single-line text input field.
- IC Number:** A single-line text input field.
- Phone Number:** A single-line text input field.
- Gender:** A dropdown menu with the placeholder text "Select Gender".
- Date of Birth:** A date picker field showing the format "dd/mm/yyyy".
- Address:** A multi-line text input field.

A small red logo is visible in the bottom right corner of the form area.

Figure 24: Adding Donor

6.3.4 Staffs

This page is where the admin specifically can view and manage other staff members registered in RedConnect. The page displays a list with the number of staff, email, full name, phone number, position in the system, and actions admin can take on the staff's profile. Actions include view, edit, and delete. Admin can also add a new staff member into the system.

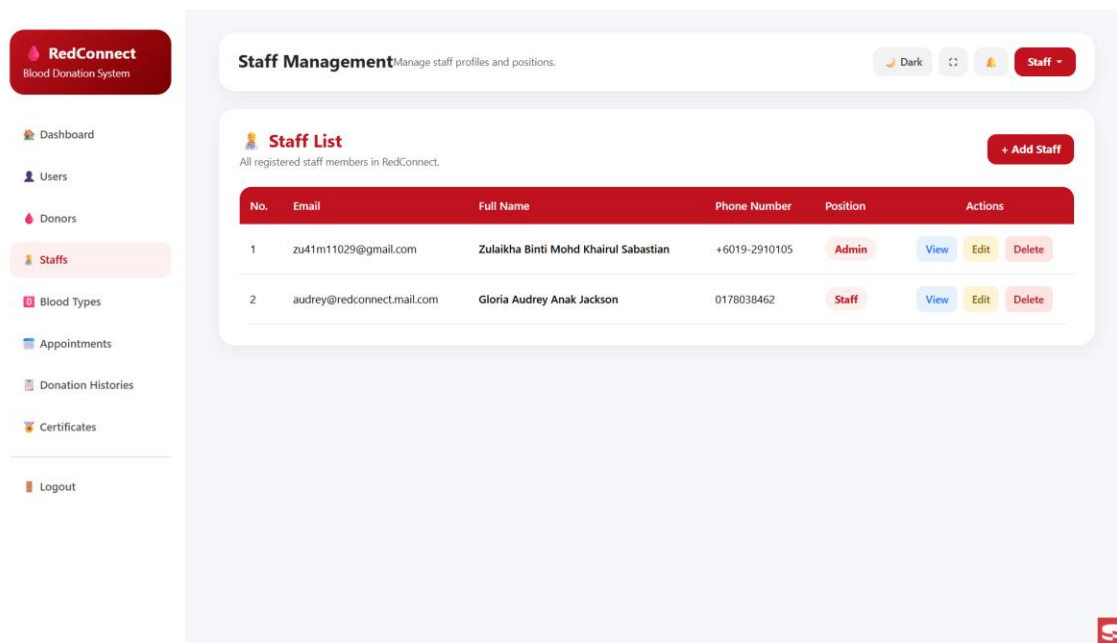


Figure 25: Staff List

When viewing a staff profile, the admin will be taken to a Staff Details page. The page displays the staff's Full Name, Email Account, Staff ID, Phone Number, and Position. Admin can also view the Related Donation Histories of the staff which includes the Donation Date, Donor ID, Appointment ID, Quantity Pack, Remarks, and Actions that can be taken such as view and edit.

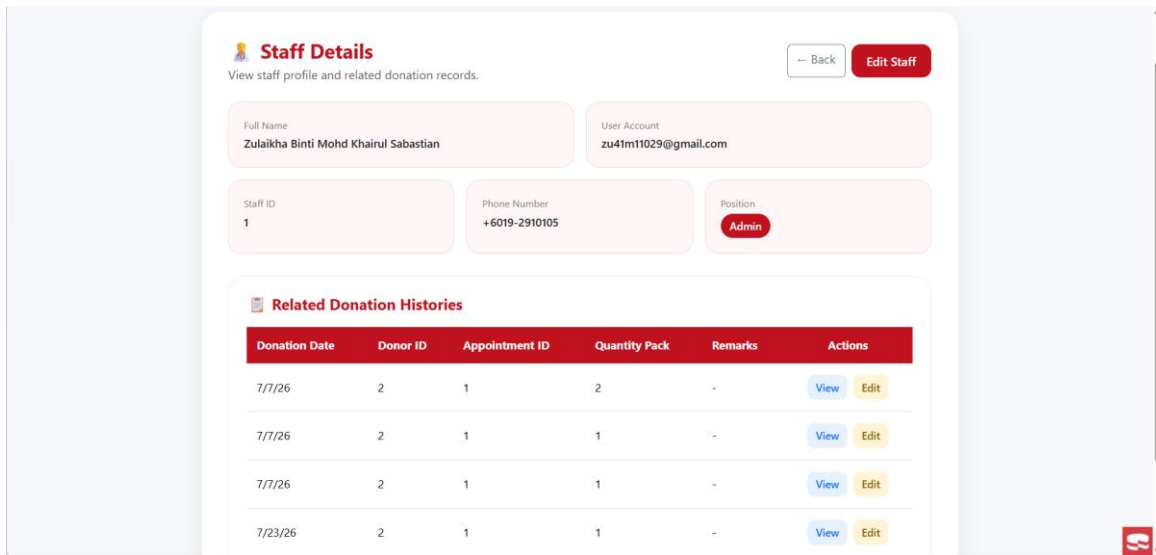


Figure 26: Viewing Staff

When editing a staff profile, admin will be able to edit the account information as well as staff information. To confirm the edit, the admin must click on the 'Update Staff' button. If not, the admin can cancel the edit. Admin has the option to delete the staff's profile as well.

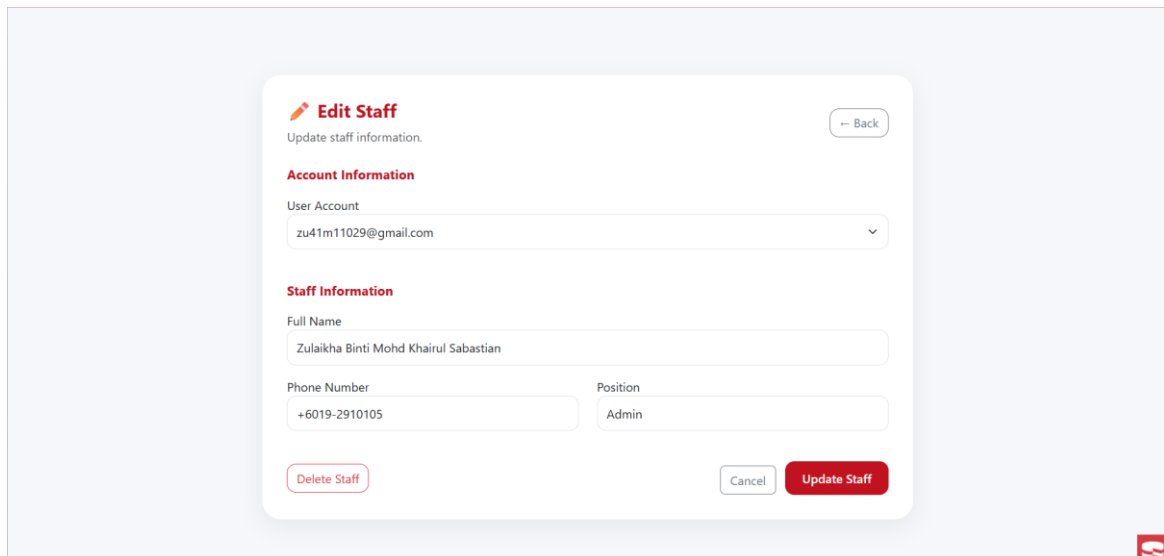
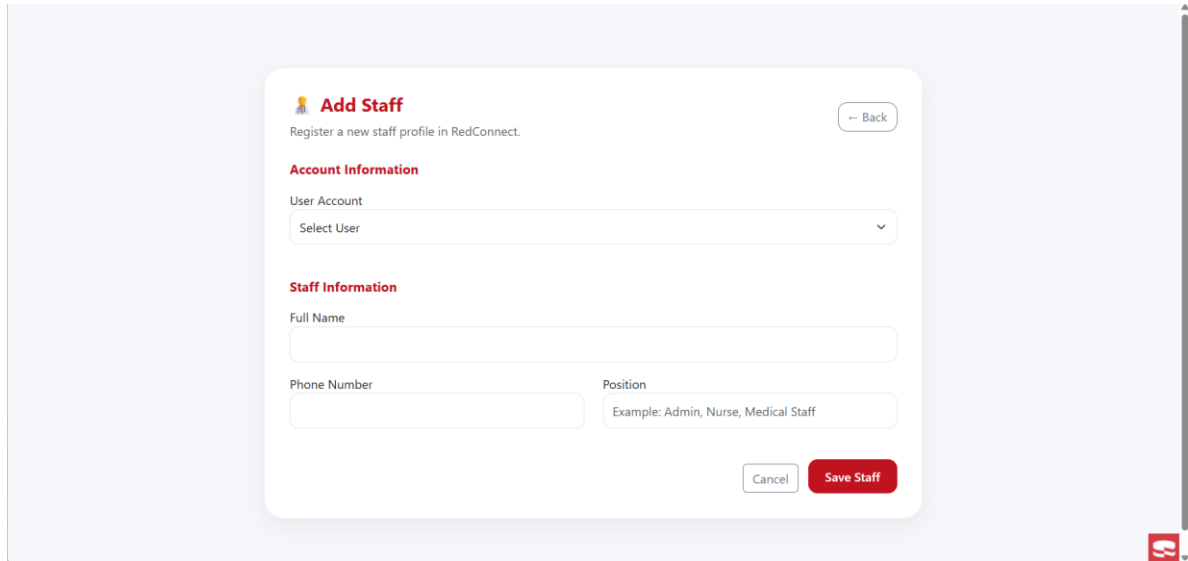


Figure 27: Editing Staff

When adding a new staff profile, the admin must input the account information and staff information. Clicking on 'Save Staff' will officially add a new staff account into the system. The admin can also cancel the action.



The screenshot shows a web application interface for adding a new staff profile. The form is titled "Add Staff" with a person icon and a subtitle "Register a new staff profile in RedConnect." A "Back" button is in the top right. The form is divided into two sections: "Account Information" and "Staff Information". Under "Account Information", there is a "User Account" label and a dropdown menu with "Select User". Under "Staff Information", there is a "Full Name" label and a text input field. Below that, "Phone Number" and "Position" are labeled with their respective input fields. The "Position" field has a placeholder text "Example: Admin, Nurse, Medical Staff". At the bottom right of the form are "Cancel" and "Save Staff" buttons. A small red logo is visible in the bottom right corner of the page.

Figure 28: Adding Staff

6.3.5 Blood Type

This page is where staff manage blood group categories used by donors. There are 8 blood types in the list: A-, A+, AB-, AB+, B-, B+, O-, and O+. Staff can view, edit, delete, or add new blood types.

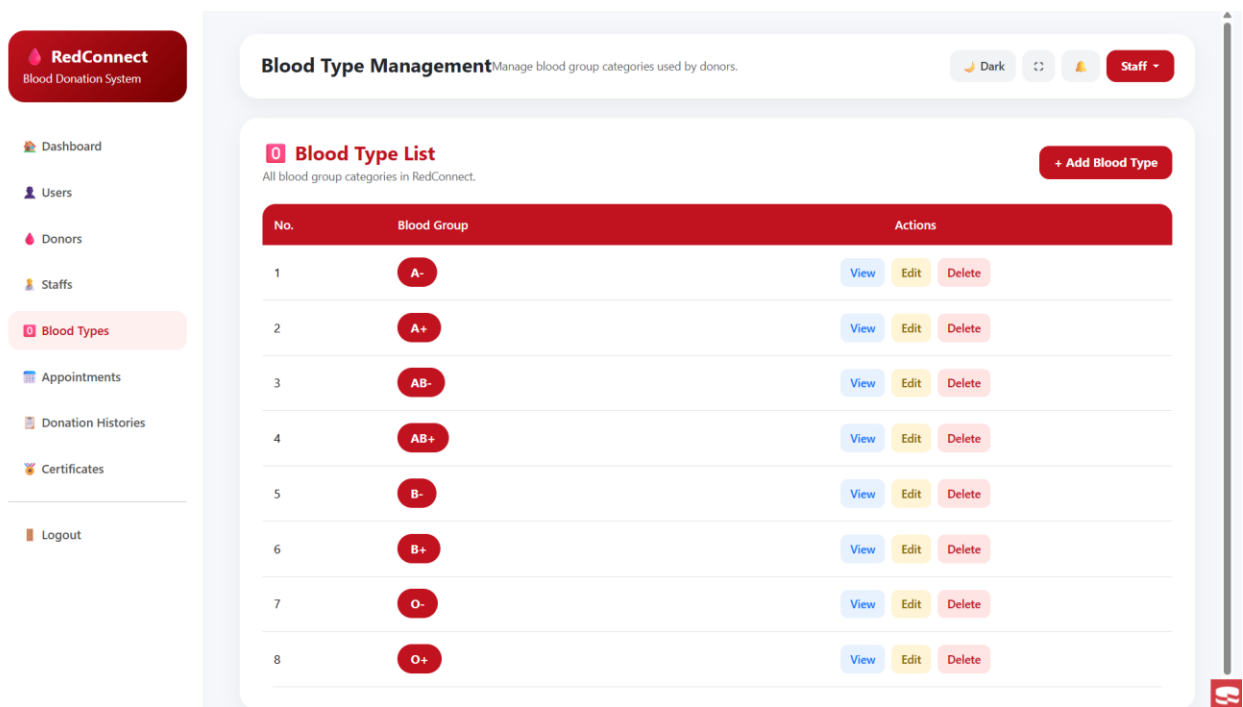


Figure 29: Blood List

When viewing the different blood types, staff will be met with the Blood Type Details page. The page will display the Blood Type ID and Blood Group. It will also display related donors to that blood type which includes the full name of the donor, their IC number, phone number, gender, date of last donation, and the action to view and edit the donor's profile.

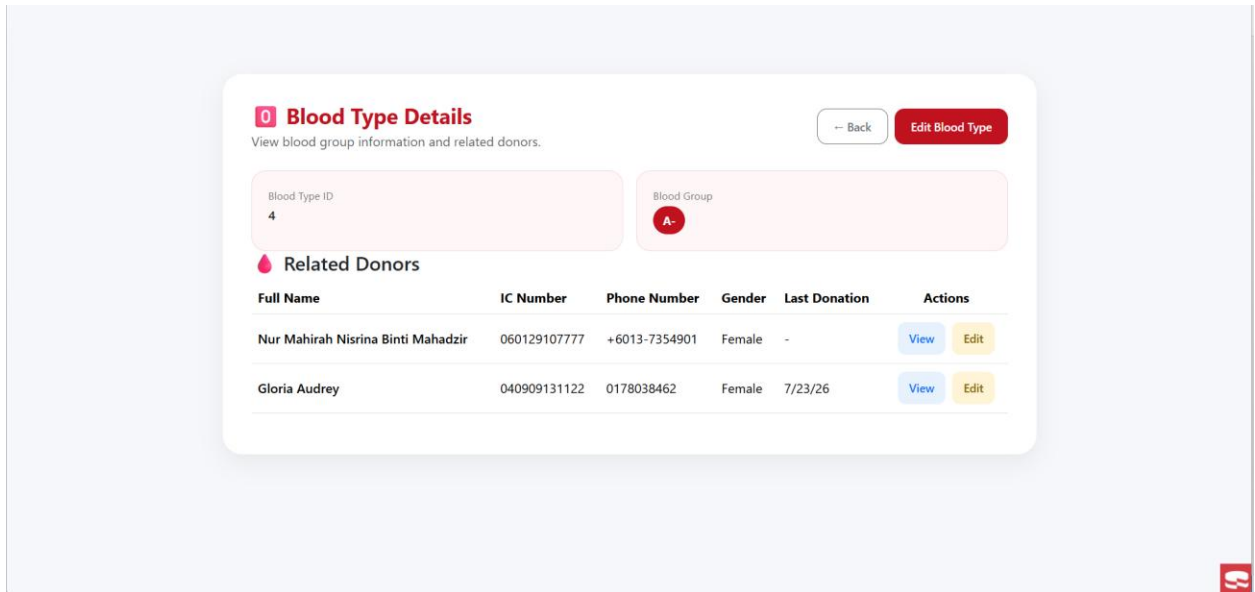


Figure 30: Viewing Blood Type

When editing the blood types, staff can change the blood group or delete it. When adding a new blood type, staff can add a new blood group into the system.

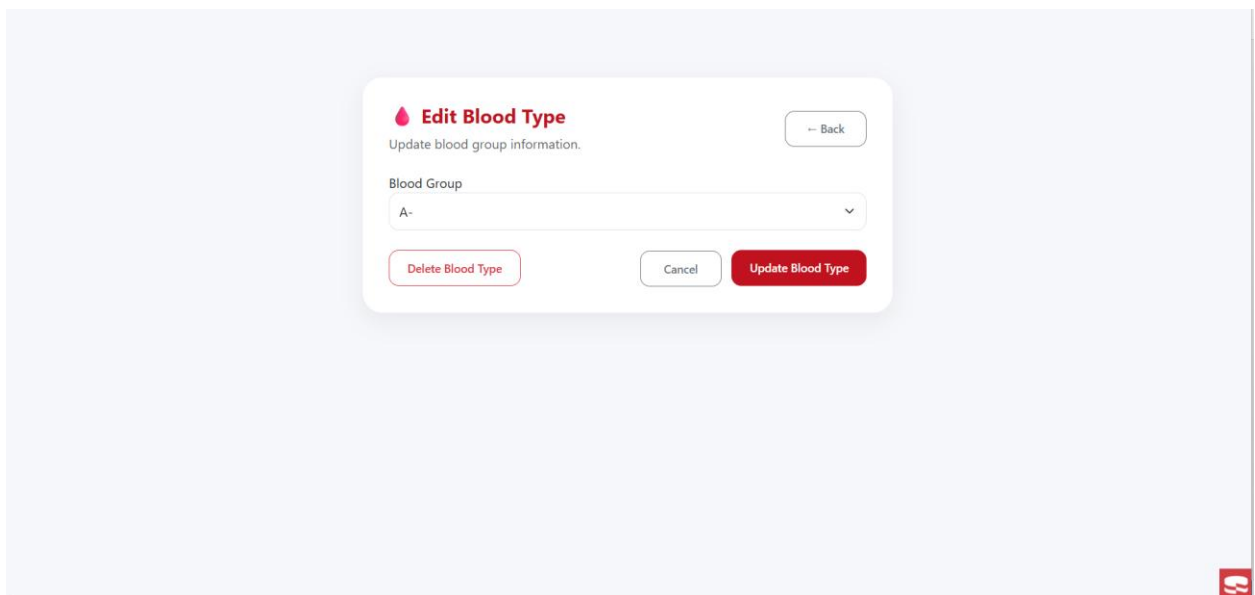


Figure 31: Editing Blood Type

Figure 32: Adding Blood Type

6.3.6 Appointments

This page is where staff can manage donor appointment bookings and approval status. The page displays an appointment list with the number of donors, donor names, appointment date, appointment time, status of appointment and actions staff can take on individual appointments such as view, edit, and delete. Staff can also add a new appointment.

No.	Donor Name	Appointment Date	Appointment Time	Status	Actions
1	Gloria Audrey	7/11/26	10:30 AM	Completed	View Edit Delete
2	Gloria Audrey	7/6/26	9:00 AM	Approved	View Edit Delete
3	MELATI	7/6/26	6:34 AM	Completed	View Edit Delete
4	MELATI	7/6/26	9:39 PM	Approved	View Edit Delete
5	Gloria Audrey	7/24/26	12:39 AM	Pending	View Edit Delete
6	Gloria Audrey	7/23/26	12:47 AM	Approved	View Edit Delete

Figure 33: Appointment List

When viewing an individual appointment, staff will be met with the Appointment Details page. This page displays the Donor's Name, Status of Appointment, Appointment ID, Appointment Date, Appointment Time, Health Declaration, and Related Donation Histories.

Appointment Details

View appointment booking and health declaration information.

← Back

Edit Appointment

Donor

Gloria Audrey

Status

Completed

Appointment ID

1

Appointment Date

7/11/26

Appointment Time

10:30 AM

Health Declaration

Healthy

Related Donation Histories

Donation Date	Donor ID	Staff ID	Quantity Pack	Remarks	Actions
7/7/26	2	1	2	-	View Edit
7/7/26	2	1	1	-	View Edit

Figure 34: Viewing Appointment

When editing an appointment, staff can edit the donor's Name, Appointment Date, Appointment Time, Health Declaration of the donor, and the Status of the appointment. To finalize the edit, staff must click on the 'Update Appointment' button or press 'Cancel' to opt out of the action. Staff can also delete the appointment from the system.

Figure 35: Editing Appointment

Staff can add new appointments to the system by inputting the necessary information such as Donor Name that is already in the system, Appointment Date, Appointment Time, and Health Declaration of donor. To successfully add the new appointment, staff must click on the 'Save Appointment' button. Staff can also cancel the action.

Figure 36: Adding Appointment

6.3.7 Donation Histories

This page is where staff can manage completed blood donation records. The page displays a Donation History List which consists of Number of Donations, Donor Names, Staff Names, Appointment Status, Donation Date, Quantity Pack, Remarks, and actions that can be taken by staff which are View, Edit, and Delete. Staff members can also Add a new Donation History.

RedConnect
Blood Donation System

Donation History Management Manage completed blood donation records.

Donation History List
All completed donation records in RedConnect.

[+ Add Donation History](#)

No.	Donor	Staff	Appointment	Donation Date	Quantity Pack	Remarks	Actions
1	Gloria Audrey	Zulaikha Binti Mohd Khairul Sabastian	Completed	7/7/26	2 pack	-	View Edit Delete
2	Gloria Audrey	Zulaikha Binti Mohd Khairul Sabastian	Completed	7/7/26	1 pack	-	View Edit Delete
3	Gloria Audrey	Zulaikha Binti Mohd Khairul Sabastian	Completed	7/7/26	1 pack	-	View Edit Delete
4	MELATI	Gloria Audrey Anak Jackson	Completed	7/6/26	1 pack	-	View Edit Delete
5	Gloria Audrey	Zulaikha Binti Mohd Khairul Sabastian	Completed	7/23/26	1 pack	-	View Edit Delete

Figure 37: Donation Histories

Viewing the Donation History will bring staff to the Donation History Details page. The page displays the Donation ID, Certificate ID, Donor Name, Staff Name, Appointment Status, Donation Date, Quantity Pack, and Remarks.

Donation History Details
View completed blood donation information.

Donation ID: 1

Certificate: RC-20260715-0005

Donor: Gloria Audrey

Staff: Zulaikha Binti Mohd Khairul Sabastian

Appointment Status: completed

Donation Date: 7/7/26

Quantity Pack: 2 Pack(s)

Remarks: -

Buttons: -- Back, Edit Record

Figure 38: Viewing Donation History

Editing the Donation History allows staff to edit the Donation information which includes the Donor's Name, Staff's Name, Appointment Status, Donation Date, Quantity Pack, and Remarks. To confirm the edit, staff must click on the 'Update Record' button. Staff can also cancel the action.

Edit Donation History
Update completed blood donation record.

Donation Information

Donor: Gloria Audrey

Staff: Zulaikha Binti Mohd Khairul Sabastian

Appointment: completed

Donation Date: 07/07/2026

Quantity Pack: 2

Remarks:

Buttons: -- Back, Delete Record, Cancel, Update Record

Figure 39: Editing Donation History

Adding a new Donation History requires staff to input the necessary Donation Information such as the Donor's Name, Staff's Name, Appointment Status, Donation Date, Quantity Pack, and Remarks. To officially add the new history, staff must click on the 'Save Donation History' button. Staff can also cancel the action.

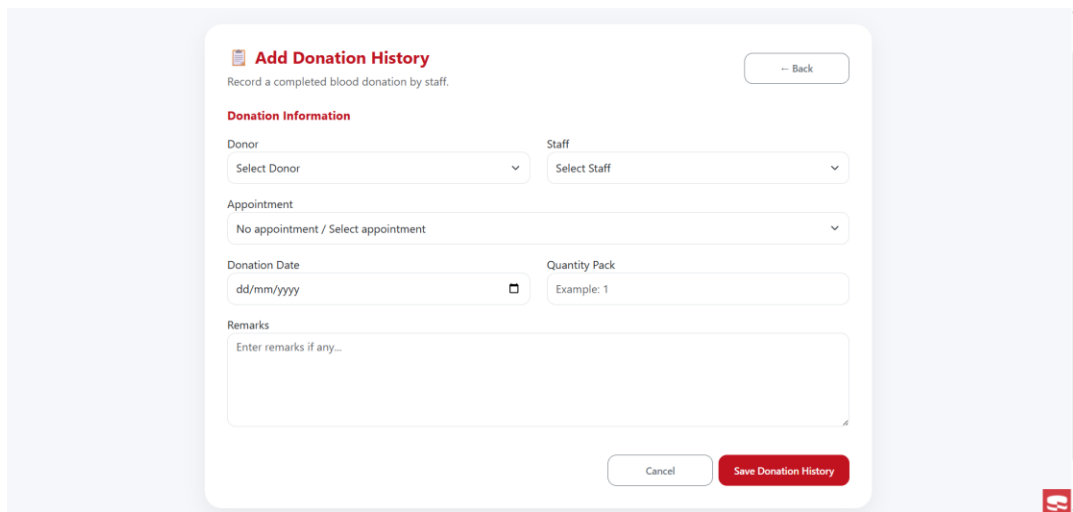
The screenshot shows a web form titled "Add Donation History" with a subtitle "Record a completed blood donation by staff." and a "Back" button. The form is divided into sections: "Donation Information" which includes dropdowns for "Donor" (labeled "Select Donor"), "Staff" (labeled "Select Staff"), and "Appointment" (labeled "No appointment / Select appointment"); "Donation Date" with a text input "dd/mm/yyyy" and a calendar icon; "Quantity Pack" with a text input "Example: 1"; and "Remarks" with a large text area labeled "Enter remarks if any...". At the bottom are "Cancel" and "Save Donation History" buttons. A small red logo is in the bottom right corner of the form area.

Figure 40: Adding Donation History

6.3.8 Certificates

This page is where staff can manage certificates generated for completed blood donations. The page displays a list of certificates generated by donors which includes the number of certificates, Donor Name, Certificate Code, Date and Time of Creation, and the actions that staff can take which are view, edit, delete, and generate Certificate.

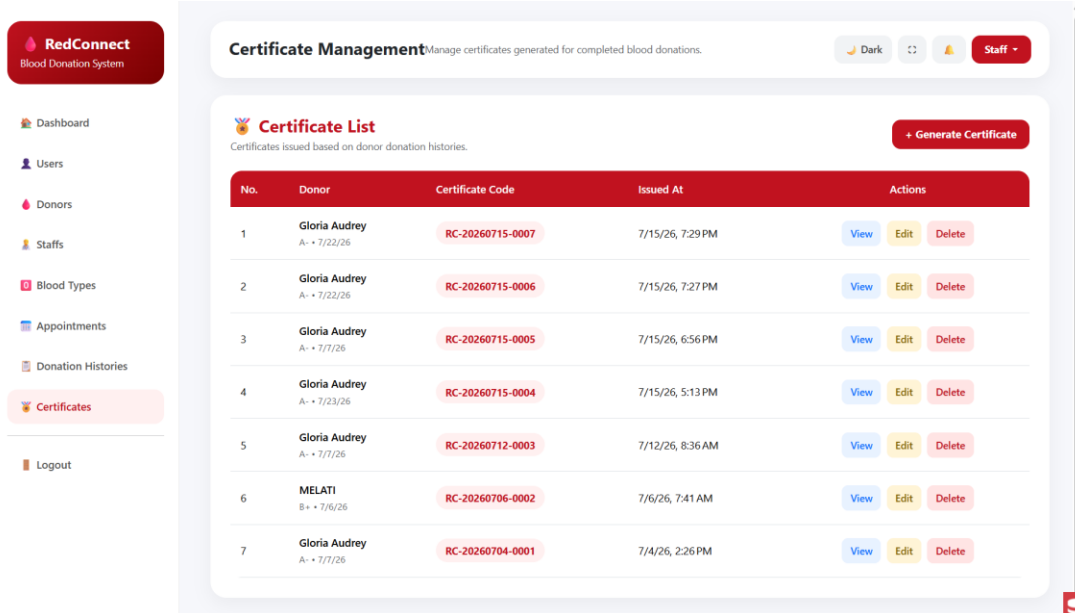


Figure 41: Certificate List

When generating a Certificate, staff will be met with the Generate Certificate page where staff can manually input the information his or herself. The Certificate Information includes Completed Donation, Certificate Code and Issued Date. To generate a Certificate with the information inputted, staff may click on the 'Generate Certificate' button. Staff can also cancel the action.

Generate Certificate
Generate a certificate for a completed blood donation.

Certificate Information

Completed Donation
-- Select Completed Donation --

Certificate Code
Generated automatically

Issued Date
15 Jul 2026

Certificate Code will be generated automatically after submission.
Issued Date will use the current date and time.
Only completed donations without an existing certificate are displayed.

[Cancel](#) [Generate Certificate](#)

Figure 42: Generating Certificate

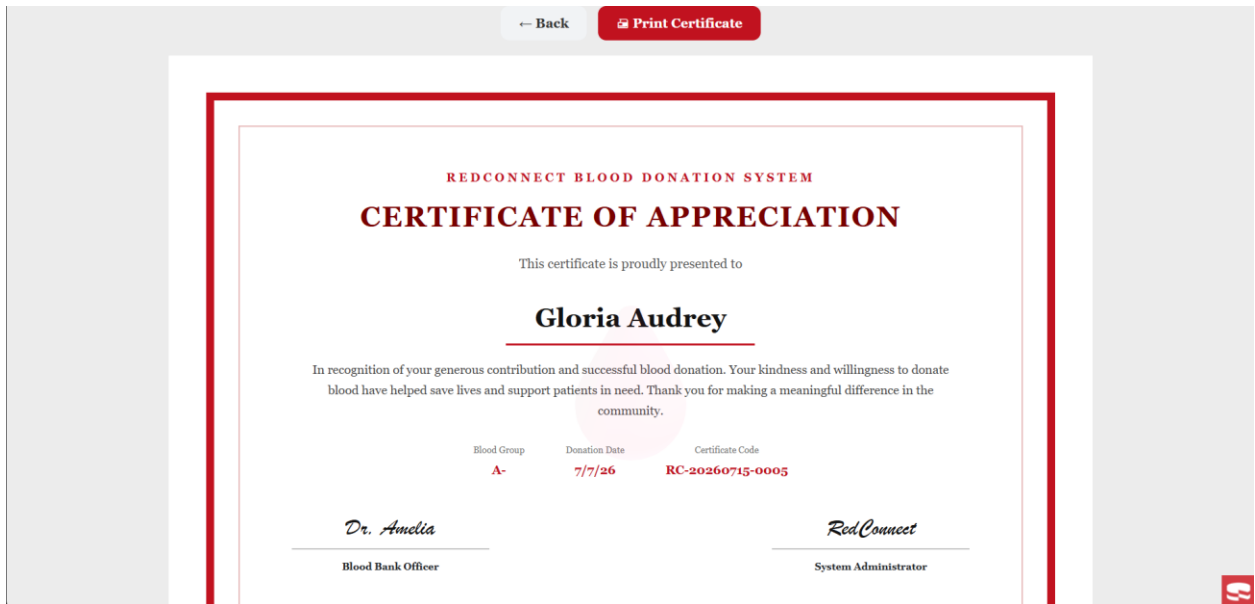


Figure 43: Certificate Generated

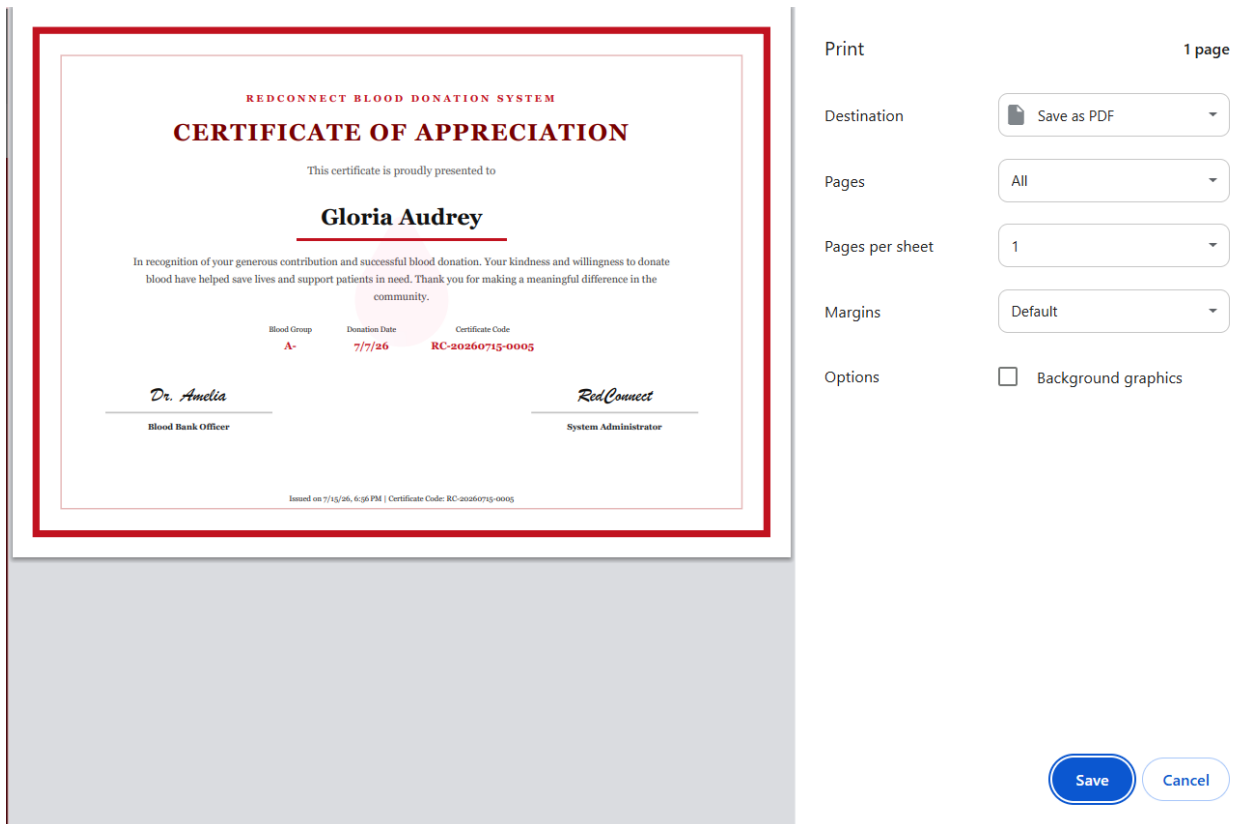


Figure 44: Printing Certificate

7.0 Workflow of Form

RedConnect system operational workflow is initiated right at the time a donor registered and login into the portal. Under motivation to contribute to the society, the donor navigates to the appointment booking modules or click on the sidebar navigation under 'Appointments' to secure the donation slot.

Implementation of the user-friendly interface, the portal specifies the preferred donation date, time and specific blood drive location. After the data details submission, the system will seamlessly record the request and stores data securely in the central database, marking the appointment as pending for administrative review.

Once recorded, the workflow transitions to the staff and system administrator portal interface, where more features are available to handle data and information submitted. Staff members review incoming requests from the donors to ensure adequate operational capacity at the blood centers. Updated information from the staff or admin portal will automatically keep the donor portal informed by logging in and monitoring their booking's progress either, the date and time requested remains pending, has been officially approved or declined for further review or marked as completed.

In order to keep strict scheduling integrity and prevent unexpected misunderstandings between the staff and donor, the system intentionally restricts donors from editing or withdrawing their finalized appointments without directly contacting the staff. Administrative control entirely delegated to staff to ensure any modifications or cancellations are handled systematically and with proper authorization.

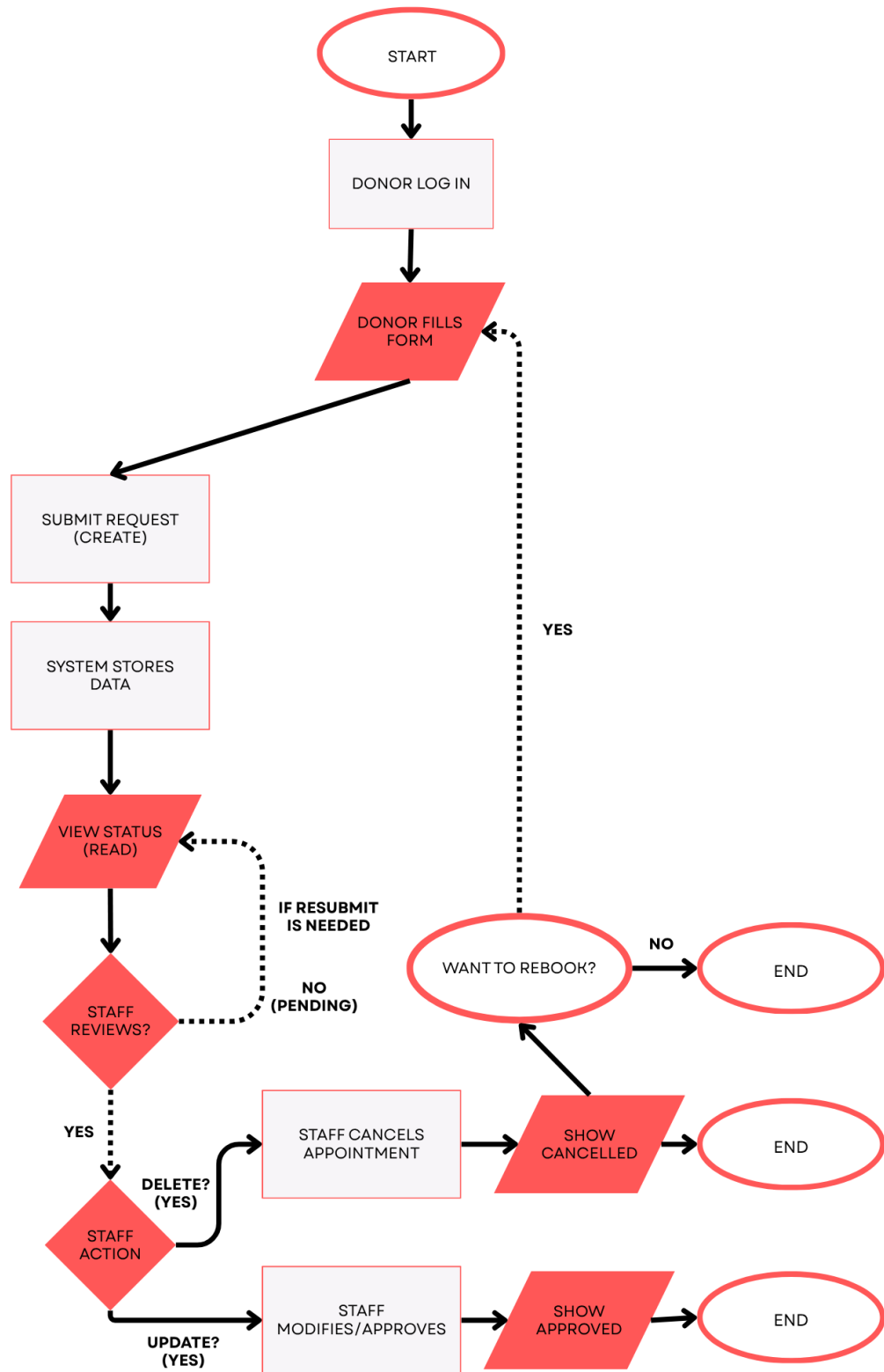


Figure 45: Flowchart of the workflow

The flowchart shown as above visually translates the operational journey into the technical CRUD (Create, Read, Update and Delete) lifecycle. It highlights the distinct accessibility levels between staff and donors. The Create operation is effectively demonstrated through the initial donor action by submitting the booking form. Sequentially, when the donor accesses their dashboard to track their request status, they are performing the Read operation.

Following the system's controlled scheduling policy, the Update and Delete operation exclusively needs to be done by the administrators. If the appointment submitted by the donor requires rescheduling or a status adjustment, the Update operation need to be executed by the staff. Lastly, the staff is given authority to perform Delete operation to safely remove records, if the booking appointment must be voided entirely due to donor ineligibility or unforeseen circumstances. This clears the separation roles between users/donors and administration to run seamlessly in unified system's operation.

8.0 Team Roles & Contributions

Team Members	Roles	Contributions
Mahirah Nisrina Binti Mahadzir	Database Designer	Designed the Entity-Relationship Diagram (ERD), developed the database schema, created tables and relationships, defined primary and foreign keys, implemented the MySQL database in CakePHP, and ensured data integrity and consistency.
Gloria Audrey Anak Jackson	Programmer	Developed the system functionalities using CakePHP, implemented CRUD operations, integrated the database with the application, tested system functions, and fixed coding errors to ensure smooth system performance.
Khadijah Sofea Binti Mohd Haidhir	System Analyst	Gathered and analysed system requirements, identified user needs, prepared the system workflow and functional specifications, evaluated system processes, and ensured the developed system met the project objectives.
Zulaikha Binti Mohd Khairul Sabastian	Project Manager	Planned and coordinated the project activities, assigned tasks to team members, monitored project progress, crossed functional boundaries by participating directly in the development lifecycle, ensured timely completion of deliverables, and facilitated communication among team members.

Table 1: Team roles and contribution

9.0 Contact Information (Support)

The Blood Donation Appointment and Donor Management System provide support information to assist users in resolving technical issues or inquiries related to the system. Users can contact the support team through the following communication channels:

Support Type	Details
System Administrator	Blood Donation Management Team
Email	support@redconnect.com
Contact Number	+60 12-345 6789
Operating Hours	Monday – Friday, 8:00 AM – 5:00 PM
Address	Faculty of Information Management, UiTM Puncak Perdana, Selangor

Table 2: Redconnect support information

Users are encouraged to contact the support team for assistance with account registration, login issues, appointment management, technical errors, or other system-related inquiries. The support team aims to provide prompt assistance to ensure the system operates smoothly and efficiently.

10.0 Conclusion & Reflection

RedConnect, Blood Donation Appointment and Donor Management System development has successfully modernized the blood donation process by striving to fully automated digital workflow, transition from the old-school manual procedures in the usual blood donation process in real-life situation. Implementation of strict workspace separation, the system effectively provides user-friendly Donor Portal tailored for donors' personal bookings alongside a comprehensive Staff Dashboard for administrative control, which gives access to data security and operational efficiency. Furthermore, the integration of an advanced UI engine using the Bootstrap 5 and Chart.js delivers pleasing and responsive data driven experience and shows appreciation mechanism, visually reinforce donors' life-saving contributions. Ultimately, RedConnect able to stand as a robust and scalable portal that effectively eliminates the gap between donors that are committed to serving communities and healthcare facilities that in need of easy access to the people eager to donate blood.

Reflecting on the development journey of RedConnect, there are several technical difficulties presented which provide invaluable learning experiences. Mastering strict Model-View-Controller (MVC) architecture within the CakePHP is the primary difficulty. It requires a steep learning curve between backend logic and frontend views understanding in ensuring data flow can be managed properly. Additionally, in ensuring real-time charts and UI components can be rendered flawlessly, seamless integration dynamic PHP data with JavaScript libraries are done to create meticulous client-side assemble. Finally, extensive mobile UX adjustments are also being made to create possibility of adapting a data-heavy dashboard for smaller or mobile screens. Iterative refinements, such as implementing hamburger menu button on the mobile view and managing independent scroll behaviours between the sidebar navigation and the content part of the page, were crucial in delivering platform-independent user experience. Coming through these obstacles by implementing continuous problem-solving behaviours significantly enhanced practical development skills and solidified the system's technical foundation.